

Karthik Garimella

501-658-4229 | kvgarimella@gmail.com | kvgarimella.github.io | [GitHub](#)

Summary

I am an ML systems researcher and a recent PhD graduate from NYU. I am looking for research engineer and scientist positions focused on machine learning systems. I am specifically interested in ML compiler/runtime infrastructure, inference systems, and performance engineering.

Education

PhD, New York University <i>Electrical and Computer Engineering</i> • Advisor: Brandon Reagen • Thesis: Designing and Optimizing Private Machine Learning Systems	2020–2026
MSc, Washington University in St. Louis <i>Computer Engineering</i> • GPA: 3.78/4.0	2018–2020
BA, Hendrix College <i>Physics (minor in Computer Science)</i> • GPA: 3.97/4.0	2013–2017

Core Technical Skills

ML systems and performance: PyTorch, JAX, Triton, cuTile, GPU memory hierarchy, Tensor Cores, Nsight Systems, Nsight Compute, `torch.compile`, IO-aware algorithms

Programming: Python, C/C++, CUDA, Rust, Go, Linux/shell, Git, LaTeX

Research areas: Deep learning, computer systems architecture, compilers, programming systems

Professional Experience

Research Assistant – New York University <i>Computing Systems Architecture, Privacy-Preserving Machine Learning</i> • Characterized, designed, and optimized machine learning systems for cryptographically secure neural network inference based on Fully Homomorphic Encryption (FHE). • HE-LRM (preprint): Co-designed embedding compression and compiler techniques to enable encrypted recommendation models. Featured in Google Security blog post. • EinHops (WAHC @ CCS 2025): developed an einsum-style programming abstraction for encrypted FHE tensors. Enabled generalized tensor contractions using a unified dataflow. • Orion (ASPLOS 2025): mapped arbitrary convolutions (e.g., any stride, padding, dilation) to efficient encrypted FHE circuits. Added support for encrypted embedding table lookups. • Cryptonite (ASPLOS 2023): identified system-level bottlenecks that prevented well-established private inference systems from scaling beyond a single inference and found parallelization strategies for the homomorphic layers. • Presented work at CRYPTO PPML, NVIDIA Research, MIT CSAIL, UC Berkeley, and Princeton.	2020–2026 New York, NY
NVIDIA Research <i>Research Intern – Programming Systems and Applications Group</i> • Co-developed a CUDASTF-based GPU framework for fully homomorphic encryption, debugging correctness issues and implementing GPU-accelerated CKKS primitives. • Designed tensor packing strategies on top of the framework and demonstrated end-to-end encrypted neural network inference on GPUs.	May 2024–Aug. 2024
NASA Jet Propulsion Laboratory <i>Software Developer Intern</i> • Built software for the Physical Oceanography Distributed Active Archive Center, supporting scientific data access and data ingestion.	May 2019–Aug. 2019
Oak Ridge National Laboratory <i>Scientific Software Developer</i> • Developed scientific software for the Climate Change Science Institute and the ORNL Distributed Active Archive Center, supporting Earth science data workflows.	Sep. 2017–May 2018

- Developed data science visualization tools for the Goddard Earth Sciences Data and Information Services Center.

Technical Projects and Writing

FlashAttention2 Forward and Backward in Triton

2026

Blog post + code

- Implemented Triton kernels for the FlashAttention2 forward and backward passes with auto-tuning and a two-pass algorithm to avoid atomic operations.
- Profiled eager PyTorch, `torch.compile`, a Triton FlashAttention2 implementation, and PyTorch FA2 with Nsight Systems.
- Showed how IO-aware tiling reduces memory pressure by avoiding materialized $N \times N$ attention matrices.
- Measured forward runtime improvements from 23.51 ms to 2.19 ms and backward improvements from 28.9 ms to 6.53 ms on an RTX 3090. Reduced memory pressure from GiBs to MiBs.
- Links: [blog](#) and [code](#).

Selected Publications

A full publication list is available on [Google Scholar](#). Selected papers:

- **2025 – Orion: A Fully Homomorphic Encryption Framework for Deep Learning**, ASPLOS. *Best Paper Award*.
- **2025 – EinHops: Einsum Notation for Expressive Homomorphic Operations on RNS-CKKS Tensors**, WAHC @ ACM CCS.
- **2023 – Characterizing and Optimizing End-to-End Systems for Private Inference**, ASPLOS.
- **2021 – A Cautionary Tale of Using Low-Degree Polynomial Activations in Privacy-Preserving Deep Learning**, PPML @ ACM CCS.
- **2020 – Attacking Vision-Based Perception in End-to-End Autonomous Driving Models**, Journal of Systems Architecture.

Awards and Leadership

Awards: 2024 MLCommons Rising Star; 2024 NYU Li Publication Award; 2022–2025 GAANN PhD Fellow; 2020 Dean's PhD Fellowship; third place, CVPR 2019 Autonomous Driving Challenge

Teaching and service: Head TA for Computing Systems Architecture (Fall 2023); Head TA for Deep Learning at NYU (Spring 2023); Lead organizer of NYU Computer Architecture Day 2024; CADathlon@ICCAD Topic Chair

Selected Talks

- **2025:** PPML @ CRYPTO – Orion: An FHE Framework with an Application to Deep Learning Recommendation Models.
- **2024:** NVIDIA Research – Packing Strategies for Fully Homomorphic Encryption.
- **2023:** MIT CSAIL Security Seminar, UC Berkeley, Princeton Computer Architecture Day – Characterizing and Optimizing End-to-End Private Inference.